

- 2

An employee of a company sells houses. He sends images of the houses to his company.
- (a)

The employee compresses the image files using lossy compression before sending them. Lossy compression reduces the size of the image files.
- Give **two** other reductions that lossy compression makes to the images.

1

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2

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[2]
- (b)

The image data is broken down into packets to be sent across a network to the company.
- The destination internet protocol (IP) address is read by a device in the network. The device uses the IP address to forward the packet towards its correct destination.
- (i)

State the name of the device that reads the destination IP address.

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[1]
- (ii)

Identify the part of the packet where the destination IP address is found.

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[1]
- (c)

The IP address has an IPv6 format.
- Give **three** characteristics of the IPv6 format.

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[3]
- (d)

An echo check is used to check for errors after the image has been transmitted.
- Draw and annotate a diagram to show how the echo check would detect any errors.

[3]
- (e)

The image data is encrypted using symmetric encryption.
- (i)

Tick (✓) **one** box to show the correct statement about symmetric encryption.

A

Encrypted text is known as plain text.

☐

B

The most secure form of encryption.

☐

C

The encryption key is shared with any device receiving the encrypted data.

☐

D

Two different keys are used.

☐

[1]
- (ii)

Give **one** other type of encryption that could be used.

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[1]